



12.1

Abelian Group (1)

12.1.1 Abelian Group: ISRO CSE 2018 | Question: 17



$(G, *)$ is an abelian group. Then

- a. $x = x^{-1}$ for any x belonging to G
- b. $x = x^2$ for any x belonging to G
- c. $(x * y)^2 = x^2 * y^2$, for any x, y belonging to G
- d. G is of finite order

isro2018 group-theory set-theory&algebra abelian-group

Answer key

12.2

Equivalence Class (1)

12.2.1 Equivalence Class: ISRO CSE 2007 | Question: 29



The set of all Equivalence Classes of a set A of Cardinality C

- A. is of cardinality 2^C
- B. have the same cardinality as A
- C. forms a partition of A
- D. is of cardinality C^2

isro2007 set-theory&algebra equivalence-class

Answer key

12.3

Functions (2)

12.3.1 Functions: ISRO CSE 2014 | Question: 50



Let A be a finite set having x elements and let B be a finite set having y elements. What is the number of distinct functions mapping B into A .

- A. x^y
- B. $2^{(x+y)}$
- C. y^x
- D. $y!/(y-x)!$

isro2014 set-theory&algebra functions combinatory

Answer key

12.3.2 Functions: ISRO-DEC2017-5



Let $f(x) = \log|x|$ and $g(x) = \sin x$. If A is the range of $f(g(x))$ and B is the range of $g(f(x))$ then $A \cap B$ is

- A. $[-1, 0]$
- B. $[-1, 0)$
- C. $[-\infty, 0]$
- D. $[-\infty, 1]$

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Answer key

12.4

Group Theory (1)

12.4.1 Group Theory: ISRO CSE 2016 | Question: 4



If $(G, .)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}, \forall a, b \in G$, then G is a/an

- A. Commutative semi group
- B. Abelian group

C. Non-abelian group

D. None of these

set-theory&algebra group-theory isro2016

Answer key

12.5

Injective (1)

12.5.1 Injective: ISRO-DEC2017-9



The function $f : [0, 3] \rightarrow [1, 29]$ defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$ is

A. injective and surjective

B. surjective but not injective

C. injective but not surjective

D. neither injective nor surjective

isrodec2017 functions injective surjective

Answer key

12.6

Relations (3)

12.6.1 Relations: ISRO CSE 2018 | Question: 56



The time complexity of computing the transitive closure of binary relation on a set of n elements is known to be

a. $O(n)$

b. $O(n * \log(n))$

c. $O(n^{\frac{3}{2}})$

d. $O(n^3)$

isro2018 set-theory&algebra relations time-complexity

Answer key

12.6.2 Relations: ISRO-DEC2017-1



Suppose A is a finite set with n elements. The number of elements and the rank of the largest equivalence relation on A are

A. $\{n, 1\}$

B. $\{n, n\}$

C. $\{n^2, 1\}$

D. $\{1, n^2\}$

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Answer key

12.6.3 Relations: ISRO-DEC2017-2



Consider the set of integers I . Let D denote "divides with an integer quotient" (e.g. $4D8$ but not $4D7$). Then D is

A. Reflexive, Not Symmetric, Transitive

B. Not Reflexive, Not Anti-symmetric, Transitive

C. Reflexive, Anti-symmetric, Transitive

D. Not Reflexive, Not Anti-symmetric, Not Transitive

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Answer key

12.7

Set Theory (5)

12.7.1 Set Theory: ISRO CSE 2011 | Question: 27



Which one of the following is true?

A. $R \cap S = (R \cup S) - [(R - S) \cup (S - R)]$

- B. $R \cup S = (R \cap S) - [(R - S) \cup (S - R)]$
 C. $R \cap S = (R \cup S) - [(R - S) \cap (S - R)]$
 D. $R \cap S = (R \cup S) \cup (R - S)$

isro2011 set-theory&algebra set-theory

Answer key 

12.7.2 Set Theory: ISRO CSE 2017 | Question: 9



The symmetric difference of sets $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 3, 5, 6, 7, 8, 9\}$ is:

- A. $\{1, 3, 5, 6, 7, 8\}$
 B. $\{2, 4, 9\}$
 C. $\{2, 4\}$
 D. $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

isro2017 set-theory&algebra set-theory

Answer key 

12.7.3 Set Theory: ISRO CSE 2020 | Question: 76



If $A = \{x, y, z\}$ and $B = \{u, v, w, x\}$, and the universe is $\{s, t, u, v, w, x, y, z\}$. Then $(A \cup \bar{B}) \cap (A \cap B)$ is equal to

- A. $\{u, v, w, x\}$
 B. $\{\}$
 C. $\{u, v, w, x, y, z\}$
 D. $\{u, v, w\}$

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Answer key 

12.7.4 Set Theory: ISRO-2013-74



The number of elements in the power set of the set $\{A, B, C\}$ is

- A. 7
 B. 8
 C. 3
 D. 4

isro2013 set-theory&algebra set-theory easy

Answer key 

12.7.5 Set Theory: ISRO-DEC2017-8



The number of elements in the power set of $\{\{1, 2\}, \{2, 1, 1\}, \{2, 1, 1, 2\}\}$ is

- A. 3
 B. 8
 C. 4
 D. 2

isrodec2017 set-theory combinatory

Answer key 

Answer Keys

12.1.1	C	12.2.1	C	12.3.1	A	12.3.2	A	12.4.1	B
12.5.1	B	12.6.1	D	12.6.2	C	12.6.3	B	12.7.1	A
12.7.2	B	12.7.3	X	12.7.4	D	12.7.5	D		